Incentives at Play:
Fee-Induced Volume on a Regulated Perpetual Futures Venue

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Abstract

Kalshi launched the first CFTC-regulated perpetual futures on 3 June 2026. Using every public trade in its BTC and ETH contracts through 20 September 2026 (26.3 million trades, $47.2 billion), its gold and silver contracts since their listing, and 283 million Binance futures trades as reference prices, I document that 39% of BTC and 48% of ETH notional is executed in resting quotes of a fixed dollar size that are taken in full by orders of the same size. CFTC filings show that self-clearing members faced no exchange fees until 24 June and net fees of zero thereafter under a rebate program filed the same day. The fixed-size flow is present from the first week of trading, has changed its size, intensity and quote-replenishment cycle in steps that hit both contracts within minutes, and is insensitive to a fee change on 18 August that moved the taker from a zero fee to 0.3 bps, roughly its gross edge. When fees went live on 24 June, venue-wide volume collapsed 80–92% within hours and returned forty hours later with ticket volume three to five times higher. On the median day the two contracts trade 33 and 61 times their open interest, against 1.3 and 1.6 times on Binance. The metals contracts, under the same program, show no fixed-size flow and track their reference market's hours. Order books recorded from 23 September show a book seven to nine times thinner than Binance's, on which fixed-size trades move the price a fifth to a half as much as other trades; neither changed when a volume incentive program ended on 13 October. The evidence is consistent with volume that exists because it is costless to produce rather than because it serves a trading purpose.

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1 Introduction

Reported trading volume is among the most widely consulted statistics a marketplace produces. Investors read it as a gauge of liquidity and of the cost of entering and exiting a position; index and data providers use volume to decide which instruments to rank, list, and include in benchmarks; and venues compete on it directly, because a contract's volume drives both its network effects and the fee revenue it earns. The same statistics feed into how trading venues are themselves valued, a question of growing salience as several such venues approach public markets. These uses give a venue and its members a private incentive to inflate the statistic, and a substantial literature treats manufactured volume, trades that transfer no beneficial ownership and convey no trading interest, as a first-order threat to the informativeness of market data (Cong et al., 2023; Aloosh and Li, 2024).

The obstacle is one of measurement. Because a manufactured trade is by construction indistinguishable at the tape from an ordinary one, establishing that volume has been inflated has required either account- or wallet-level data that identify the counterparties (Victor and Weintraud, 2021; Sirolly et al., 2025), or statistical signatures of the trade record that are suggestive but not dispositive (Cong et al., 2023; Aloosh and Li, 2024). Neither approach observes the incentive to produce volume; each infers manufactured trading from its footprint rather than from the payoff that would motivate it.

This paper studies a setting in which that incentive is observed rather than inferred. Kalshi, a Designated Contract Market regulated by the Commodity Futures Trading Commission (CFTC), launched the first CFTC-regulated perpetual futures on 3 June 2026. As a regulated exchange, Kalshi self-certifies every fee schedule, rebate program, and trading incentive to the CFTC with a stated effective date, so the pecuniary reward to producing a marginal trade can be reconstructed from the public regulatory record and dated to the hour. This makes the venue clean in one dimension and opaque in another. It is clean because the incentive to trade is written down and changes on known dates; it is opaque because the public trade tape carries no account identifiers, so the defining legal element of a wash sale, common ownership or prearrangement of the two sides, cannot be observed directly. The empirical strategy turns on the first feature to work around the second. Rather than attempt to establish ownership, I ask a question the data can answer: does a given component of reported volume respond to dated changes in fees and rebates the way independent, profit-seeking trading would, or does it behave as volume produced because it is costless to generate?

The answer is sharp. Across the contracts' entire history I identify a flow of trades executed in resting quotes of a fixed dollar size that are taken in full by orders of the same size, and this flow accounts for 39% of bitcoin and 48% of ethereum notional. It is present in the first week of trading, and its dollar size, its intensity, and its quote-replenishment cycle are reconfigured in discrete steps that strike both contracts within minutes of one another, the signature of a single automated process rather than of independent traders. It is insensitive to a fee change on 18 August that moved its taker side from a zero net fee to a cost of roughly its entire gross edge, whereas identifiable arbitrage flow on the same venue responded to a simultaneous fee cut as profit-seeking trading should. When exchange fees first went live on 24 June, venue-wide

volume fell by 80–92% within hours and returned some forty hours later with the fixed-size flow several times larger. Gold and silver contracts, listed later under the same rebate program, exhibit no such flow and track their reference market's trading hours as closely as the reference market itself, ruling out an explanation in the venue's matching engine, its interface, or its membership. Order books recorded after the sample show the flow resting on a book seven to nine times thinner than Binance's and barely moving the venue's own price. Throughout, the money changing hands on these trades is a few thousand dollars a day against hundreds of millions in notional. The evidence is consistent with volume that exists because producing it is costless, not because it serves a trading purpose. All figures are computed from the raw trade tape.[1]

The paper makes five contributions. First, it assembles the full institutional timeline. The perpetual contracts traded with no exchange fee for three weeks; a fee schedule went live at 10:00 ET on 24 June; a "Temporary Perpetual Fee Rebate Program" filed the same day returned all net fees to self-clearing members; an update effective 18 August changed the rebate so that takers net a 0.3 bps fee and makers a 0.3 bps credit; and a further update effective 16 September extended the program to metals and added, for the first time, a clause excluding wash trades from rebates. Between two members, exchange fees therefore netted to zero from 24 June onwards, and a member trading with itself paid nothing.

Second, it documents the fixed-size flow over the contracts' whole life rather than a two-week window. The flow is present from the first week of June. Its dollar size was changed on 10, 18 and 24 August in both contracts, its share of volume doubled on 10 August and on 26 June, and the cycle at which its quotes are replenished dropped from about 520–700 ms to about 118 ms at 06:07 UTC on 25 August, in both contracts within the same few minutes and 27 hours after the sizes had changed. Whatever produces this flow is a single configurable process, and it was reconfigured twice in two days.

Third, it uses the dated fee events as natural experiments. The 24 June switch-on of fees is followed within hours by a collapse of venue-wide volume of 80–92%, and by a return on 26 June with ticket volume three to five times its pre-event level and the ETH ticket doubled from \$5,000 to \$10,000. The 18 August change moved the ticket taker from a zero net fee to 0.3 bps, which is roughly its entire gross edge; neither its volume, its selectivity nor its markouts changed, while volume from identifiable latency arbitrageurs rose several-fold in response to a simultaneous cut in list fees. Across all regimes the money changing hands on the fixed-size trades is a few thousand dollars a day on hundreds of millions of notional: whatever the flow is for, it is not for trading profit on either side.

Fourth, it uses the gold and silver contracts listed on 10 September as a within-venue comparison. They are covered by the same rebate program but show no fixed-size flow, their size concentration is an order of magnitude below the crypto contracts, and their volume falls by 86–96% when COMEX is closed, in line with the reference market. The crypto anomaly is

[1] The fixed-size pattern first drew practitioner attention in September 2026 (CoinDesk, 2026); a contemporaneous pseudonymous working paper documented it over a two-week window and applied the detectors of the wash-trading literature (OctopusTakopi, 2026). The present paper differs in covering the contracts' full history, reconstructing the fee and incentive schedule from the regulatory record, and testing the flow against the dated changes in that schedule rather than against statistical benchmarks.

therefore not a property of the venue's matching engine, its API or its member base as a whole.

Fifth, it adds order-book evidence. Kalshi's public API shows the current order book but keeps no history of it, so I recorded the Kalshi and Binance books for all four contracts from 23 September. Kalshi's crypto books are seven to nine times thinner than Binance's and 120 to 350 times costlier to sweep for $250,000, so the turnover documented in Section 7 would cost takers who pay to cross the book 0.4–2.6% of open interest every day. Fixed-size trades move Kalshi's own mid a fifth to a half as much as other trades, and neither the flow nor the book changed when Kalshi ended a volume-based incentive program on 13 October.

The paper relates to the literature on wash trading in crypto markets, which has relied either on account- or wallet-level data (Victor and Weintraud, 2021; Sirolly et al., 2025) or on statistical signatures of the trade tape (Cong et al., 2023; Aloosh and Li, 2024). It adds a setting where the incentive to produce volume is written down in public filings and changes on known dates, which allows behavior to be tested against the incentive rather than against a statistical benchmark. It also relates to work on exchange fee design and maker–taker pricing (Colliard and Foucault, 2012; Malinova and Park, 2015), where the concern is usually distortion of order choice rather than of the volume statistic itself.

2 Institutional setting

Kalshi listed bitcoin and ethereum perpetual futures on 3 June 2026, followed by XRP on 10 June, further crypto contracts through the summer, and gold and silver on 10 September. The contracts trade continuously, are cleared by Kalshi's own clearing house (Kalshi Klear), and settle funding every eight hours. Members trade either as self-clearing members (SCMs) or through a futures commission merchant (FCM).

Table 1 lists every filing bearing on trading costs, assembled from the CFTC's rule-filing index for organization KEX and from Kalshi's regulatory notices page. Three points matter for what follows.

No fees at launch. The first perpetual fee schedule was filed on 8 June and, by an exchange notice, implemented at 10:00 ET on 24 June. For three weeks after launch no exchange fee applied to perpetual trades at all. The schedule that then took effect charged takers 12.0 to 2.6 bps and makers 5.0 to 0.6 bps, depending on 30-day trailing volume across perpetuals and event contracts; maker tiers could alternatively be reached by holding a minimum share of the venue's total maker volume, up to 10% for the top tier.

Full rebate from the same day. On 24 June Kalshi also filed the Temporary Perpetual Fee Rebate Program, under which all SCMs receive monthly rebates of "all net taker and maker fees" in every perpetual market. The filing is not simultaneous with the fee, however. Its cover letter is dated 24 June but was date-stamped by the CFTC on 25 June and posted to Kalshi's notices page that day, and its terms set the effective date at 9 July, "retroactive to the self-certification filing date" but with "no Rebates ... earned unless this Program goes into effect".[2]

[2]The rebate program's cover letter is dated 24 June but was submitted through the CFTC portal and date-stamped 25 June (filing 61360), and Kalshi posted it to its regulatory notices page on 25 June. Its terms set the effective date at "on or after 5:00 PM ET on July 9, 2026," with an effective period that "may be retroactive to the self-certification filing date," and "no Rebates will be earned unless this Program goes into effect."

For roughly forty hours, from 10:00 ET on 24 June, members thus faced the full fee schedule with no rebate in force and no public certainty that one would apply. Once the rebate was in effect, net exchange fees for members were zero on both sides, retroactive to 24 June. An update filed on 4 August and effective 18 August changed the terms so that "taker fees will be rebated down to 0.3 bps" and "maker fees will be rebated such that makers net 0.3 bps"; the sum across the two sides of a trade remains zero. A member on both sides of its own trade still pays nothing. The 2 September update, effective 16 September, added metals at 0.018 bps, excluded Kalshi affiliates, and for the first time excluded from rebates any fees on transactions "resulting from, or subject to inquiry or investigation for potential, self-matching, wash trading, pre-arranged trading". The June and August versions contained no such clause.

Other incentives. Kalshi amended its 2025 Market Maker Program on 3 June to add "tailored liquidity conditions and benefits" for perpetuals; the appendices setting out those conditions and benefits were filed with a request for confidential treatment under FOIA Exemption 4 and are not public. A referral program (effective 7 August), a clearing-house program paying 7% a year on open positions (effective 28 August, capped at $7 million), and a retail deposit-and-trading reward program (filed 15 September) were also filed; none of these pays for perpetual volume as such at any scale, and none predates the fixed-size flow. Two exchange-wide programs also applied. A Volume Incentive Program paid members without a market-maker agreement a share of a fixed reward per market in proportion to their volume, and its terms allowed perpetual markets to be eligible; Kalshi filed its termination on 28 September. A Liquidity Provider Program, effective from 18 May 2026 to 1 December 2027, pays members with a market-maker agreement for quoting obligations, with its selection terms filed as a confidential appendix. Section 8 returns to both.

3 Data

Kalshi publishes every perpetual trade through its public API with a price, a contract count, the taker side and a microsecond timestamp; there is no account field. I collected every trade in the BTC and ETH contracts from listing on 3 June to 20 September 2026 (18.89 million and 7.45 million trades, $27.1 and $20.1 billion of notional), and in the gold and silver contracts from listing on 10 September (250,000 and 134,000 trades, $150 million each). Cumulative notional matches the exchange's own reported lifetime volume for each contract to within rounding, and no trade identifier appears twice. One BTC contract is 0.0001 BTC and one ETH contract is 0.001 ETH, so a trade's dollar notional is price times count times the contract multiplier.

Reference prices come from Binance USD-M futures aggregated trades for BTCUSDT, ETHUSDT, XAUUSDT and XAGUSDT, downloaded from Binance's public archive for the same days (283 million trades in total). Each Kalshi trade is matched to the last Binance trade price at or before its timestamp, and to the prices 100 ms before and 1, 5, 30 and 300 seconds after. Markouts are signed in the taker's favor and expressed in basis points; they are gross of fees and funding. Because they use last-trade prices rather than quotes, they carry noise at short horizons and could bias the taker's edge in either direction by a few hundredths of a basis point, which is the same order as the net figures in Section 5.2. No order book history

Table 1: Fee and incentive filings for Kalshi perpetual futures

Filed	Effective	Content
4 May	18 May	Liquidity Provider Program (all markets): members with a market-maker agreement paid for quoting obligations, up to \$50,000 a week per series; selection terms confidential; runs to 1 Dec 2027.
3 Jun	3 Jun	BTC perp lists. Market Maker Program amended for perps; terms confidential.
4 Jun	18 Jun	Fee Consideration Program: prior volume on Hyperliquid, Coinbase, Binance, KCEX counts toward initial fee tier.
8 Jun	24 Jun 10:00 ET	First perp fee schedule: takers 12.0–2.6 bps, makers 5.0–0.6 bps by 30-day volume; maker tiers also by share of venue maker volume.
24 Jun filed, 25 Jun posted	9 Jul, retroactive	Temporary Perpetual Fee Rebate Program: all net maker and taker fees rebated to SCMs, all perp markets. Effective 9 Jul, retroactive to the filing date; no rebate earned until it takes effect.
4 Aug	18 Aug 17:00 ET	Rebate update: takers net 0.3 bps, makers net −0.3 bps. New fee schedule: takers 4.0–1.0 bps, makers 2.0–0 bps.
14 Aug	28 Aug	Clearing-house Perpetual Interest Program: 7% p.a. on $\min(\text{OI}/2, \text{margin})$, \$7m cap.
2 Sep	16 Sep 17:00 ET	Rebate update: metals at 0.018 bps; affiliates excluded; wash and self-matched trades excluded from rebates. Metals fee schedule.
10 Sep	10 Sep	Gold and silver perps list.
28 Sep	13 Oct (earliest)	Volume Incentive Program terminated. It paid members without a market-maker agreement a share of a fixed per-market reward in proportion to volume; perpetuals exempt from its price range.

Sources: CFTC rule filings 61122, 61074, 61360, 62335, 63563, 63564 and Kalshi Klear's 14 August filing; Kalshi exchange notice of 24 June 2026; Kalshi's Liquidity Provider Program filing of 4 May 2026 and Volume Incentive Program termination filing of 28 September 2026. Effective dates are those stated in the filings; implementation is "upon exchange notice", and for the 18 August and 16 September changes no implementation notice was found, so the stated earliest dates are used.

is public, so resting sizes and the replenishment cycle are inferred from trade timestamps, and open interest is taken from single snapshots of the exchange's API. Section 8 uses order books I recorded myself from 23 September, after the end of the trade sample. The metals sample is eleven days. Event contracts, which are most of Kalshi's business, are not covered, and nothing in this paper identifies a person or a firm.

A *ticket* is defined per market-day, without reference to any prior list of sizes: for every round dollar amount K (a multiple of \$100 above \$500) the window $[K - 15, K + 1]$ collects trades whose notional is the largest whole number of contracts that fits under K; any K whose window carries at least 5% of the day's notional is a ticket for that day. The procedure is given no list of sizes and recovers every ticket, past and present, from the tape alone. Because the limit is in dollars, the contract count of a ticket trade moves with the price, which is why tests built on exact contract sizes miss it.

4 The fixed-size flow

4.1 Extent

Over the full sample, ticket trades are 39% of BTC notional (\$10.6 billion) and 48% of ETH notional (\$9.7 billion). Figure 1 shows daily notional split into ticket and other volume. The

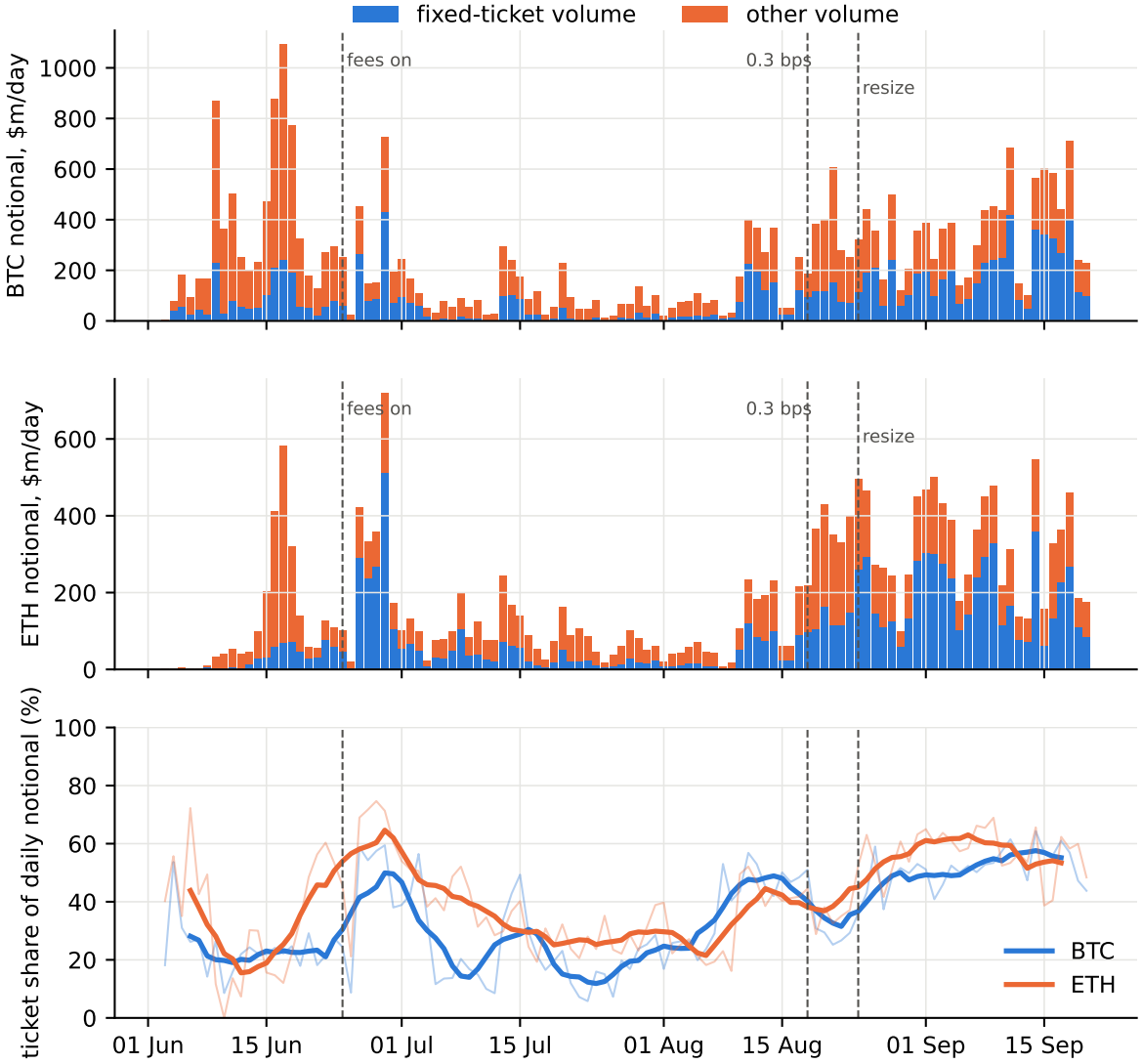


Figure 1: Daily notional in the Kalshi BTC and ETH perpetuals, 3 June – 20 September 2026, split into trades at a fixed dollar ticket (blue) and all other trades (orange), and, in the bottom panel, the ticket share of daily notional for both contracts (daily values faint, seven-day centred mean bold). Dashed lines: fee schedule live (24 June), rebate changed to 0.3 bps (18 August), ticket sizes changed (24 August).

flow is present from the first days of trading; on 3–8 June, when daily volume was in the low millions, $1,000, $5,000 and $10,000 tickets already carried 35–72% of ETH notional. In gold and silver the same procedure finds tickets on a handful of launch days only, carrying 1% of notional overall.

Figure 2 compares size concentration across venues: for each market-day with at least 500 trades, the largest share of notional inside any 1%-wide window of trade size. The median is 37% for Kalshi ETH and 24% for Kalshi BTC; it is 1% or below for every Binance market, including gold and silver, and 7% and 12% for Kalshi gold and silver. BTC's median is lower than ETH's not because its flow is less concentrated but because BTC carries two tickets at once, and a 1% window captures only the larger of them; the two together carry about the same share as the single ETH ticket. The Kalshi metals contracts sit between the crypto contracts

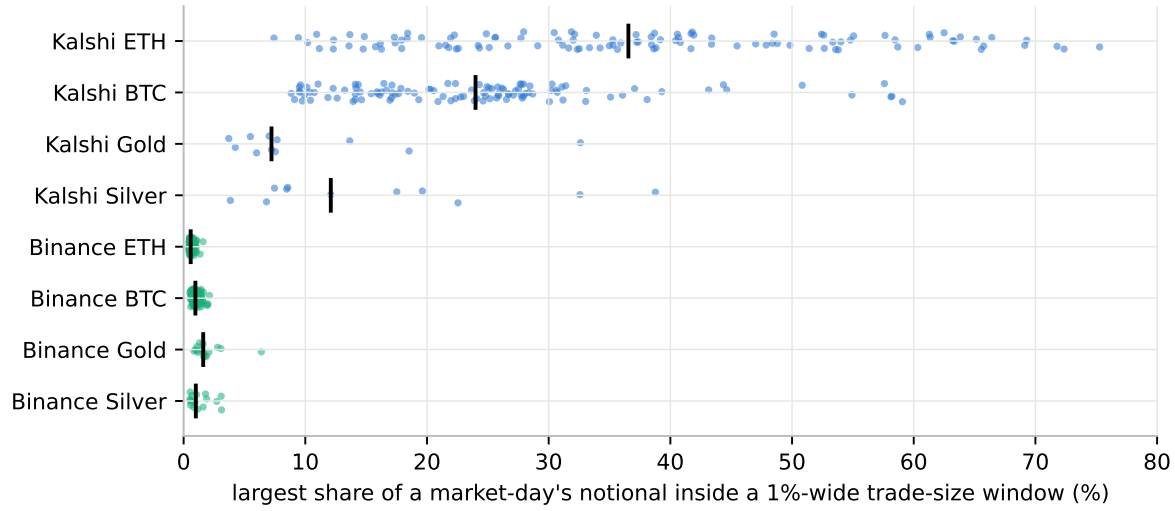


Figure 2: Dollar-size concentration per market-day. Each dot is one market-day with at least 500 trades; the bar is the median. Kalshi contracts in blue, Binance USD-M futures in the same assets in green.

and Binance, but their concentration comes from small launch-day samples rather than from a persistent ticket.

4.2 Share of volume

Figure 3 puts the extent of the suspect flow in one place. Because the ticket share has risen through the contracts' life, a single full-sample figure would understate what the venue's reported volume consists of today. In the three fee-free weeks after launch, tickets were 23% of notional in both contracts. Under the full rebate they were 37% of BTC and 48% of ETH. From the 10 August scale-up to the end of the sample they were 49% and 53%, and in the first twenty days of September 55% and 60%, or 57% of the two contracts combined. BTC and ETH are more than 85% of all perpetual volume on the venue. The same construction appears at lower shares in the smaller contracts: over 1–20 September, tickets carry 29% of HYPE notional (a $2,400 ticket), 16% of SOL ($3,500) and 11% of XRP, so the figure for the perpetual complex as a whole is a few points below the BTC and ETH figures.

Two qualifications apply. First, the ticket share is the share of volume that fails the economic tests of this paper (insensitive to fees, leaving no position, produced by one configurable process) and not an estimate of wash trading in the legal sense, which the public data cannot deliver; it is best read as an upper bound on same-owner or pre-arranged volume and as a point estimate of volume that does not measure trading interest. Second, a further layer of volume depends on the tickets without being part of them. The latency-arbitrage flow that hits Kalshi quotes within 100 ms of a Binance move of more than 1 bps, which is a further 24% of ETH and 14% of BTC notional in September, would largely not exist without stale fixed-size quotes to pick off. On that reading, roughly 85% of reported ETH volume is either ticket flow or flow induced by it. Perpetuals are in turn a minority of Kalshi's business: on 20 September the exchange reported $1.9 billion of total volume, of which the BTC and ETH perpetuals were about $0.36 billion, so the ticket flow is on the order of 10–15% of the exchange's total reported volume on

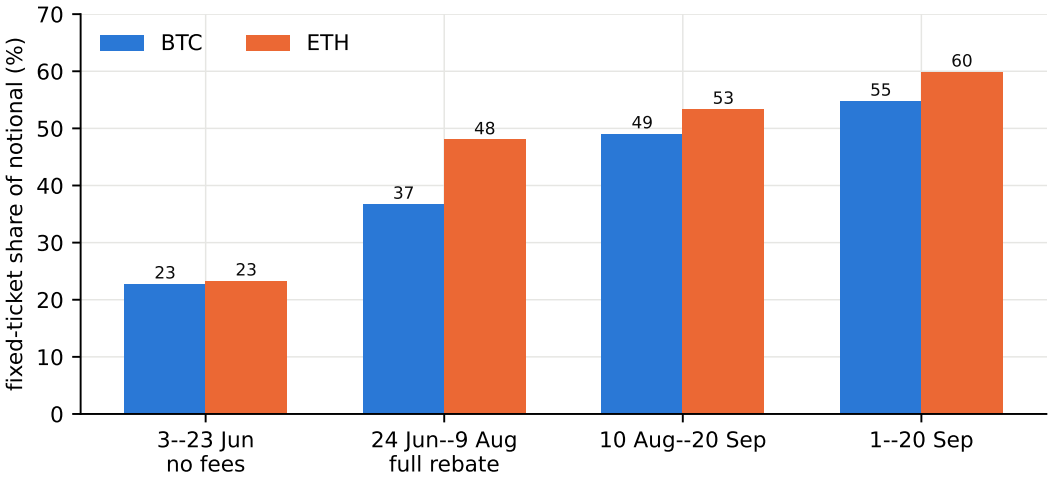


Figure 3: Share of notional executed at a fixed dollar ticket, by contract and window. Tickets are identified per market-day as in Section 3; windows follow the fee regimes of Table 1, with the last window showing the most recent three weeks.

a typical day. No comparable analysis exists for the event contracts that make up the rest.

Table 2 states the pattern of Figure 3 as a regression. On the panel of contract-hours, the ticket share of hourly notional is regressed on indicators for three later periods, with the fee-free launch weeks omitted, on hour-of-day fixed effects, and on log Binance notional, log realized volatility and a weekend indicator, separately for each contract. The three later windows here are disjoint, so the August window ends on 31 August. All three period effects are positive and the Wald test rejects their joint absence in both contracts.[3] The effects grow over time: about 15 to 22 percentage points under the full rebate, 21 to 25 points in late August and 33 to 39 points in September, and the September step above August is precise in both contracts. Because the periods are consecutive blocks of one time series, I also run a randomization test that holds the period lengths fixed, shifts all boundaries jointly to a random date, and re-estimates the model 1,000 times; the p-value is the share of draws that produce a coefficient at least as large as the observed one. The August and September effects pass this test in both contracts (BTC: 0.01 and 0.02; ETH: 0.06 and 0.001). The full-rebate effect does not, with a randomization p-value of about 0.2 in both contracts, and its unweighted estimate is close to zero (0.01 in BTC, 0.05 in ETH). That effect is therefore carried by the high-volume days in the week after the rebate filing rather than by a broad shift across the period; July hours look much like June hours. The negative coefficient on Binance notional restates the elasticity result of Section 5.3: other flow rises with the market and ticket flow rises less, so the ticket share falls in busy hours.

4.3 A single configurable process

Three features of the flow change in steps, and the steps coincide across the two contracts. The dollar size of the ETH ticket was \$5,000 from mid-June, \$10,000 on 26–29 June, back to \$5,000 from 30 June, \$4,000 from 10 August, \$4,500 from 18 August (about 06:00 UTC, fifteen hours

[3] Newey–West standard errors with a seven-day lag deliver qualitatively similar results. They are larger than the day-clustered errors by 30 to 50 percent for the period effects, and every effect that is significant at 1 percent on clustered errors remains so.

Table 2: Ticket share of hourly notional by fee period

	BTC	ETH
24 Jun–9 Aug, full rebate	0.145***	0.219***
	(0.044)	(0.058)
10–31 Aug	0.212***	0.250***
	(0.028)	(0.046)
1–20 Sep	0.333***	0.387***
	(0.020)	(0.042)
log Binance notional	-0.070***	-0.175***
	(0.020)	(0.032)
log realized volatility	0.038	0.137***
	(0.040)	(0.052)
Weekend	-0.049*	-0.033
	(0.025)	(0.037)
Hour-of-day fixed effects	yes	yes
Observations (contract-hours)	2,598	2,596
R^2	0.51	0.42
Wald test, all period effects zero (p-value)	<0.001	<0.001

Weighted least squares on contract-hours, 3 June to 20 September 2026, with hourly notional as weights. The dependent variable is the share of the hour's notional executed at a fixed dollar ticket. The omitted period is 3–23 June, when no exchange fee applied (weighted mean share 0.23 in BTC and 0.24 in ETH). Standard errors clustered by day (110 clusters) in parentheses. *, **, ***: significance at 10, 5 and 1 percent on the clustered errors.

before the fee change that day could take effect) and $5,500 from 24 August. BTC moved from $5,000 to a pair of $4,000 and $2,100 on 10 August and to $5,000 and $2,500 on 24 August; On 24 August the last ETH trade at the old size is stamped 03:07:22.98 UTC and the first at the new size 03:07:24.71; the first BTC trade at a new size is stamped 03:07:14.73, so the two contracts switched within ten seconds of each other. Daily ticket notional rose about tenfold on 10 August in both contracts (from $16 million to $155 million a day in BTC and from $9 million to $87 million in ETH, comparing 1–9 August with 10–14 August) and three- to fivefold on 26 June.

The third feature is timing. For consecutive trades of the same ticket on the same side of the book, the first percentile of the inter-trade gap, which estimates the time the quoting process needs to replace a quote that has just been taken, was between about 520 and 730 ms in ETH on 40 of the 51 days from 22 June to 23 August with more than 2,000 such gaps, and between 116 and 135 ms on every such day from 25 August (Figure 4). BTC shows the same two levels, with more noise from days on which two tickets are hit in one sweep. The change is sharp. In ETH the last hour on the old cycle (05:00–06:00 UTC on 25 August) has a first-percentile gap of 546 ms and the first hour on the new one (06:00–07:00) of 120 ms; at one-minute resolution the switch occurs at 06:07 UTC, with no gap under 300 ms in the preceding hour and twenty in that minute (Figure 5). BTC switches between 06:06 and 06:12 UTC. Figure 4 shows the two regimes as distributions: the ticket curves are flat at zero up to the cycle length and turn sharply at 520 ms in the two weeks before the change and at 118 ms in the two weeks after, while the curve for all other trades in the same market rises from 20 to 25 percent at one millisecond

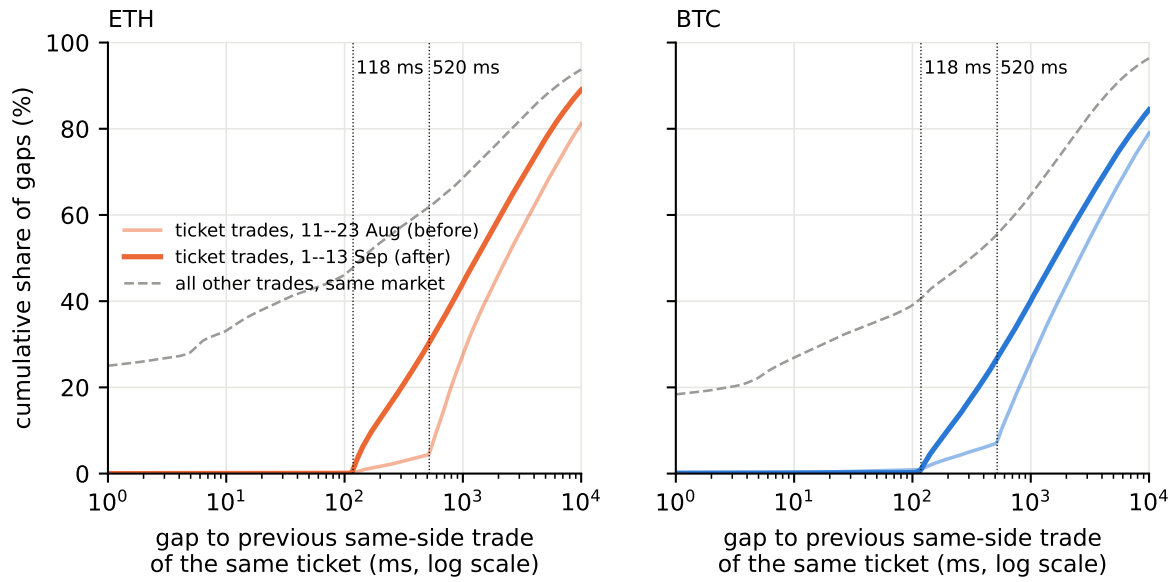


Figure 4: Replenishment cycle of the ticket quotes. Cumulative distribution of the gap between consecutive trades of the same ticket on the same side of the book, for the two weeks before the change (11–23 August) and the two weeks after (1–13 September), against the same gap for all other trades in the same market over both windows.

and has no edge anywhere. The small mass of the before curves between 118 and 520 ms comes from days on which two tickets were active and from sweeps that take two resting quotes of one size in a single burst. The ticket sizes had changed 27 hours earlier, at 03:07 UTC on 24 August, and the cycle stayed at about 520 ms through that change; the two reconfigurations are separate events, each of which reached both contracts within minutes. Neither coincides with anything in Kalshi's public API changelog, which has no entry between 20 and 27 August and none concerning order handling or rate limits until 10 September, and I found no status-page incident or public discussion of either date. The 24-hour windows either side of the cycle change show no change in the taker's markouts (0.27 bps at five seconds in ETH before and after) and a modest rise in the BTC ticket share of volume from 33% to 46%. Non-ticket trades in the same markets show no such floor: 36–68% of their same-side gaps are under 100 ms in every month. Independent makers would not share one cycle time, and would not change it together with their order sizes on the same day.

4.4 No position is left behind

Ticket buys and sells are balanced to within a fraction of a percent (50.3% buys in BTC, 50.1% in ETH), and the absolute net signed ticket flow within an hour is 12% of gross in BTC and 6% in ETH. Non-ticket flow is similarly balanced at the hourly level (11% and 10%), so this statistic alone does not separate the two; but combined with the exchange's reported open interest it says that ticket volume does not accumulate into positions. On 21 September Kalshi's API reported open interest of $11.1 million in BTC against 24-hour volume of $667 million, and $9.2 million in ETH against $621 million, so open interest turns over 60 and 67 times a day; across all 23 perpetual contracts the ratio is 42. On Binance USD-M futures over 15–20

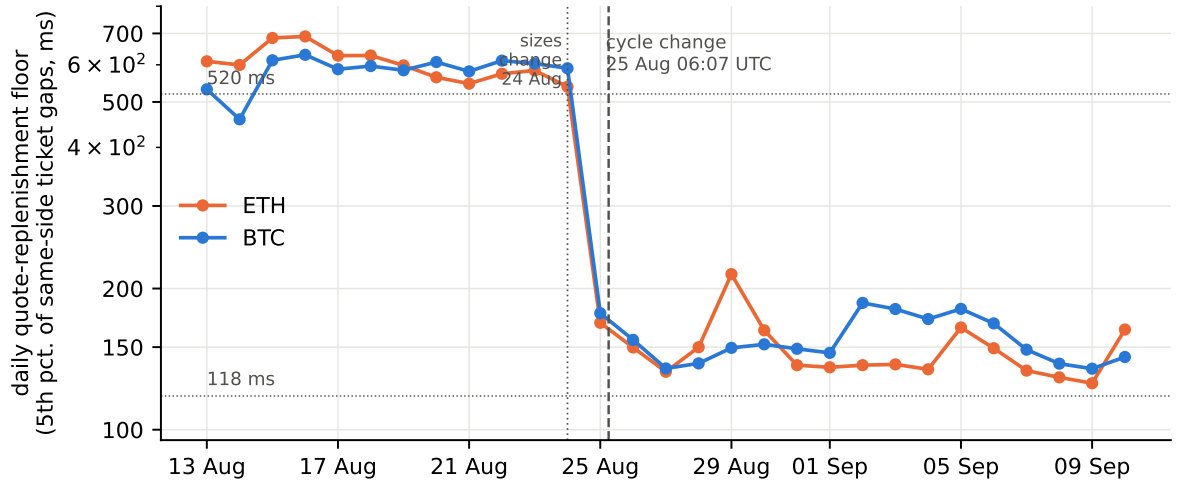


Figure 5: The quote-replenishment floor, by day. Each dot is one day's fifth percentile of the gap between consecutive trades of the same ticket on the same side of the book, a robust estimate of the shortest time the quoting process takes to replace a quote. Both contracts sit at about 520–650 ms through 24 August and drop together to about 150 ms on 25 August; the ticket-size change of 24 August (dotted) leaves the floor unchanged. Days with more than 5,000 same-side ticket gaps, 13 August–10 September 2026. Dotted horizontal lines at 520 and 118 ms.

September, daily volume was 1.4 times open interest in BTC and 1.7 times in ETH, using the exchange's published open-interest series.

5 Fee events

5.1 24 June: fees switched on

The exact times matter here, so I set them out first. The perpetual fee schedule filed on 8 June was implemented "as of 10:00 AM ET" on 24 June by an exchange notice of that date, which is 14:00 UTC. From that instant takers owed up to 12 bps and makers up to 5 bps. The rebate that would return those fees was filed the same day but was not effective the same day: its cover letter is dated 24 June, it was date-stamped by the CFTC and posted to Kalshi's notices page on 25 June, and its own terms set the effective date at 9 July, retroactive to the filing date but earning "no Rebates . . . unless this Program goes into effect". For about forty hours, then, members faced a live, unrebated fee, and on the morning the fee began the rebate had not yet even been filed.

Figure 6 shows what followed, at hourly resolution. Through 23 June and the morning of 24 June the venue runs at $10–40 million per hour and at 2–5% of Binance, with a pre-fee median of 3.1% of Binance per hour. Within hours of 14:00 UTC the flow thins, and from about 18:00 UTC on 24 June through all of 25 June it is close to dead: the quietest hour, 07:00 UTC on 25 June, traded essentially nothing in either contract and registered a Kalshi-to-Binance share of zero, against roughly $10 million and 3% in the same hour a day earlier. Over the whole of 25 June BTC traded $23 million and ETH $21 million, against $296 million and $109 million on 23 June, and ticket volume fell to $2.0 million and $4.5 million. The collapse is venue-wide, ticket and non-ticket flow alike, and the bottom panel shows it is not a market-wide lull: Binance

BTC+ETH volume was in fact near its June high on 24–25 June, so Kalshi's share, which divides that out, falls by more than a factor of ten. Trading resumed abruptly at about 08:00 UTC on 26 June and ramped to a peak of $87 million in the 14:00 UTC hour, above anything in the pre-event days; over the day BTC traded $453 million and ETH $423 million, ticket volume was $267 million and $292 million, three and five times its 23 June level, and the ETH ticket had doubled from $5,000 to $10,000, holding 69–75% of ETH volume for the next four days.

The sequence lines up with the fee timeline rather than with the market. The drop begins in the hours after fees go live and deepens overnight, which is when a fee of up to 12 bps on volume that carries a sub-basis-point edge makes that volume unambiguously loss-making; and the rebound begins on the morning after 25 June, the day the rebate filing became public and its retroactive design could be read. An operator producing volume for its own sake would stop the moment it became costly and restart the moment the cost was credibly promised to be refunded, which is what the two turning points show. That the flow returned larger, with a doubled ticket, is consistent with the same operator resuming under a confirmed zero-fee regime rather than with independent traders drifting back.

Two alternative readings deserve a word. A market-wide halt is ruled out by the bottom panel, since Binance was busy throughout. A pure technical outage is a weaker fit than it first looks: trades print in most hours of 25 June rather than stopping cleanly, the withdrawal is graded over the evening of 24 June rather than instantaneous, and the one clean gap, 07:00–09:00 UTC on 25 June, is a two-hour window in both contracts that recurs as an ad hoc maintenance pattern later in the sample. I cannot exclude that a maintenance window contributed to part of the 25 June trough, and the exchange did file notice of an ad hoc maintenance period on 28 June; but an outage explains neither the timing of the onset relative to the fee nor the timing of the recovery relative to the rebate filing, and it would have to have begun within hours of the fee and ended within hours of the rebate becoming public. The joint pattern is what fee sensitivity looks like, and a coincident outage is not needed to account for it.

5.2 18 August: the taker starts paying

The exact implementation hour of the 18 August change is not documented; the filing gives 17:00 ET as the earliest time, and I use it. From 18 August the ticket taker pays a net 0.3 bps where it paid nothing before, and the maker receives 0.3 bps where it received nothing. The sum is unchanged, but the split now determines which side loses money. Table 3 reports the taker's gross markout against Binance by fee regime. In BTC it is 0.30–0.40 bps at 30 seconds in every regime; in ETH it is 0.37–0.43 bps. At 5 seconds it is 0.27–0.39 bps. Under the third regime the taker's net edge after fees is therefore between zero and 0.1 bps, and the maker's loss net of rebate is the same amount with the opposite sign.

An independent taker facing a new 0.3 bps fee on trades with a 0.3–0.4 bps gross edge should become more selective and trade less. In a window from 11 to 23 August, excluding 18 August itself and stopping before the 24 August resizing, BTC ticket volume was $124 million a day before and $107 million after; ETH ticket volume rose from $75 million to $130 million. The Binance price move in the 100 ms before a ticket trade, a measure of how selectively the taker times its orders, was 0.10–0.19 bps throughout; five-second markouts were unchanged at 0.24–

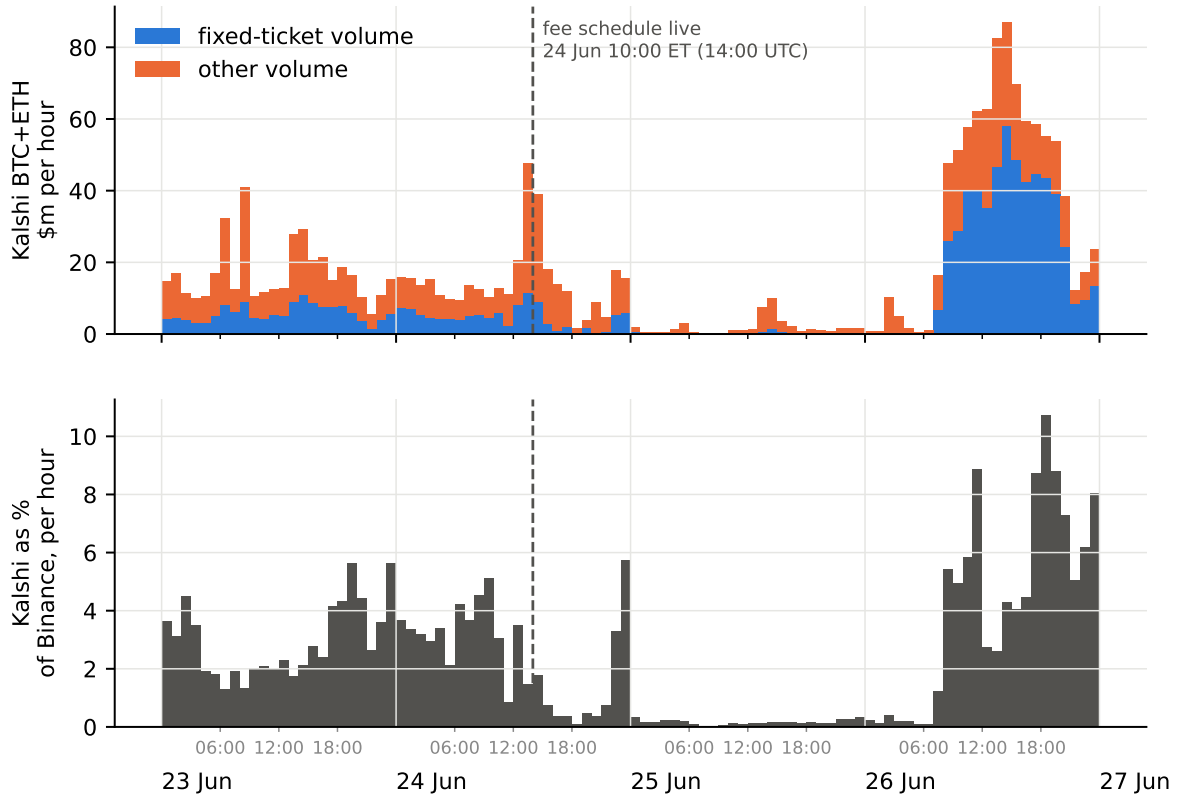


Figure 6: The 24 June switch-on of fees, at hourly resolution. Top: Kalshi BTC+ETH notional per hour, split into trades at a fixed dollar ticket (blue) and all other trades (orange). Bottom: the same Kalshi volume as a percentage of Binance BTC+ETH USD-M volume in the same hour, which nets out any market-wide movement in trading. Dashed line: perpetual fee schedule live, 24 June 10:00 ET (14:00 UTC). Sample: 23–26 June 2026.

0.29 bps. The flow that *did* respond was non-ticket flow arriving within 100 ms of a Binance move of more than 1 bps, which rose four- to six-fold after 18 August; the same filing cut list taker fees for non-members from 12.0–2.6 bps to 4.0–1.0 bps, and this is what a price-sensitive population looks like when its costs fall.

The test does not reject an independent ticket taker: at break-even it has no reason to change. What it establishes is that the economics are negligible. At 0.3–0.4 bps on $180 million a day, the gross transfer between the two sides of the ticket trades is $5,000–7,000 a day, and net of the rebate split it is at most about $1,500 a day either way. Before 18 August the maker lost the entire markout on every fill, roughly $2,000–2,500 a day, for eight weeks, with no rebate to offset it and, as far as the public filings show, no other payment.

5.3 Sensitivity to market activity

Volume that serves a trading purpose rises when markets are active. Table 4 regresses log hourly Kalshi notional on log hourly Binance notional in the same asset, separately for ticket and non-ticket flow and by regime, with standard errors clustered by day. In the launch regime nothing on Kalshi tracks Binance, which reflects the launch ramp rather than the flow's nature. From 24 June onwards non-ticket flow has an elasticity of 0.74–0.94 and a correlation of 0.58–0.76 with Binance volume; ticket flow has an elasticity of 0.36–0.86 and a correlation of 0.30–0.49.

Table 3: Ticket flow by fee regime

	Regime	Days	Vol/day ($m)	Ticket/day ($m)	Share	Ticket taker markout (bps)		
						5 s	30 s	300 s
BTC	3–23 Jun, no fees	22	354	81	0.23	0.38	0.32	0.36
	24 Jun–18 Aug, full rebate	55	133	54	0.41	0.27	0.30	0.34
	19 Aug–20 Sep, taker 0.3 bps	33	364	178	0.49	0.34	0.40	0.45
ETH	3–23 Jun, no fees	22	111	27	0.24	0.39	0.43	0.47
	24 Jun–18 Aug, full rebate	55	122	58	0.47	0.36	0.40	0.40
	19 Aug–20 Sep, taker 0.3 bps	33	331	181	0.55	0.33	0.37	0.32

Markouts are the signed change in the Binance USD-M price from the moment of the Kalshi trade, in the taker's favor, weighted by notional, gross of fees and funding.

The ETH ticket is the least responsive (0.36, correlation 0.30, in the third regime) and BTC's the most. Whether these gaps are statistically distinguishable is tested in Appendix A, which stacks the two flow types in one regression per contract and sample and interacts the Binance variable with a ticket indicator, so that the interaction is the difference in elasticities with a standard error from the joint covariance. For ETH the difference is negative and significant at 1 percent in the full sample and in both post-fee periods, and it widens over time: -0.27 under the full rebate and -0.38 under the 0.3 bps taker fee for Binance volume, and -0.60 for realized volatility in the last period. For BTC the picture is weaker than the point estimates suggest. Over the full sample and under the full rebate the difference is small and not significant; only from 18 August is the BTC ticket significantly less responsive than other BTC flow (-0.17 for volume, -0.34 for volatility, both at 1 percent). The BTC ticket therefore behaved like ordinary flow for most of the summer and detached from the market in the final regime, while the ETH ticket was detached throughout. In both contracts the gap is largest in the period in which the ticket share is highest. The ticket flow is not indifferent to the market, but where it is distinguishable it is less tied to it than everything else on the venue, and its hour-of-day profile is flatter (coefficient of variation 0.25 for the ETH ticket against 0.32 for other ETH flow and 0.39 for Binance).

6 Metals as a within-venue comparison

Gold and silver perpetuals were listed on 10 September under the same rebate program (at a lower rate from 16 September) and the same matching engine. If fixed-dollar quoting were a property of the venue's technology, its API, or its members in general, it should appear in these contracts too. It does not. No persistent ticket appears; the largest concentration on any gold day after the launch weekend is below 7%.

Metals also allow a test that crypto does not: the reference market closes. COMEX gold and silver futures do not trade on Saturdays, on Sundays before 18:00 ET, or in the daily 17:00–18:00 ET break, while Kalshi's contracts trade continuously. Figure 7 shows volume per hour when COMEX is closed as a share of the open-hours level. Kalshi gold trades at 14% of its open-hours rate when COMEX is closed and silver at 4%; Binance's XAUUSDT and XAGUSDT trade at 10% and 22%. The Kalshi metals contracts behave like their reference

Table 4: Elasticity of hourly Kalshi notional to hourly Binance notional

	Regime	Ticket flow		Non-ticket flow	
		β (s.e.)	corr.	β (s.e.)	corr.
BTC	no fees	0.04 (0.19)	0.02	0.06 (0.24)	0.03
	full rebate	0.86 (0.15)	0.40	0.94 (0.07)	0.66
	taker 0.3 bps	0.61 (0.06)	0.49	0.78 (0.03)	0.64
ETH	no fees	−0.36 (0.24)	−0.16	−0.15 (0.31)	−0.06
	full rebate	0.58 (0.14)	0.32	0.85 (0.10)	0.58
	taker 0.3 bps	0.36 (0.10)	0.30	0.74 (0.04)	0.76

OLS of $\ln V_h^{\text{Kalshi}}$ on $\ln V_h^{\text{Binance}}$ over hours h with positive Kalshi volume; standard errors clustered by day.

market. Kalshi's crypto contracts trade at 32% and 45% of their weekday-hours rate during the same clock hours, against 38% and 47% on Binance: there is nothing to close in crypto, and the profile is ordinary. This is a placebo in the sense that matters: a venue whose volume is manufactured by its own systems or by an exchange-wide arrangement would not produce metals volume that switches off with COMEX.

7 Volume relative to open interest

The preceding sections argue that much of the reported volume in the crypto perpetuals is manufactured rather than traded. A natural objection is that these are young, small contracts, and that a young, small contract may simply trade actively relative to its size, so high volume need not indicate anything untoward. This section addresses that objection with a metric that draws on a variable the ticket construction never uses—open interest—and that should be the sharpest place for manufactured volume to show itself. Turnover, the ratio of a day's traded notional to open interest, benchmarks volume against the size of the positions the market actually carries. It is the natural sanity check here because a round-trip that opens and closes a position within the day adds to volume while leaving open interest unchanged, and trades that are self-matched or pre-arranged net to no position and so add volume and no open interest at all. If the fixed-size flow is not building positions, turnover on the crypto contracts should stand far above that of an established venue trading the same underlyings; and the metals contracts, where Section 6 finds no such flow, should not.

I take Binance USD-M futures in the same four underlyings as the yardstick. Open interest for the Kalshi perpetuals is taken from the exchange's margin-market candlestick history (hourly, in dollars); for Binance it is taken from the exchange's published open-interest metrics (five-minute, in dollars). Volume is each venue's own reported traded notional. Both venues report volume as a single count of each trade in dollars, verified against the raw trade records: Kalshi candlestick volume matches the deduplicated trade tape to within 0.05%, and Binance matches its aggregated-trade archive exactly, so the two turnover series are on the same basis.[4]

[4]Open interest is single-sided and dollar-denominated on both venues. Turnover uses each day's mean open interest; using end-of-day open interest instead leaves the picture unchanged. The sample and dating follow the rest of the paper (BTC and ETH from 3 June, gold and silver from 10 September, through 20 September 2026). The metals series are eleven trading days.

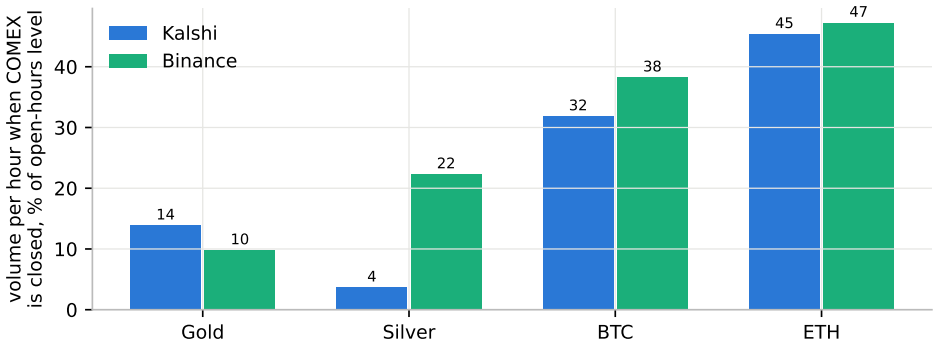


Figure 7: Volume per hour during hours when COMEX metals futures are closed (Saturday, Sunday before 18:00 ET, 17:00–18:00 ET daily), as a percentage of volume per hour during COMEX trading hours, 10–20 September 2026.

Table 5 and Figure 8 report the result. Over the sample, median daily volume is 33 times open interest for the Kalshi BTC perpetual and 61 times for ETH, against 1.3 and 1.6 on Binance—a factor of 25 to 38. Kalshi turnover exceeds Binance's on every one of the 110 trading days in both contracts. The gold contract behaves entirely differently: it turns over 5.1 times its open interest on Kalshi against 4.7 on Binance, a ratio of 1.1, and the two series track each other day to day, including the shared troughs when the COMEX floor is closed over weekends. Silver sits between the two cases, at 19 against 3.5. The crypto contracts thus carry positions that are minuscule relative to the volume printed against them, in a way the metals contracts under the same rebate program do not, and the reference venue does not.

Read as a sanity check, the exercise cuts the way the volume analysis predicts. The excess is specific to volume measured against positions, and to the crypto contracts: gold, the within-venue placebo, matches the reference market almost exactly, so nothing about Kalshi's clearing, its API, or its member base produces inflated turnover on its own. Because open interest is orthogonal to the fixed-dollar-ticket construction of Section 4 and is not inflated by round-trip trading, the agreement of the two approaches is genuine corroboration rather than a restatement. Turnover of tens of times open interest per day is not what position-taking looks like; it is what volume that leaves no position looks like. Two independent approaches—the response of the flow to dated fee changes, and its volume relative to the positions it leaves—therefore point the same way. What neither can settle is who runs the process that produces this volume, the question Section 9 takes up after a look at the order book behind the volume.

8 The order book behind the volume

Kalshi's public API shows the current order book but provides no history of it, so past states of the book cannot be retrieved (Section 3). From 23 September 2026 I therefore recorded it myself: Kalshi's full order book for the four perpetuals, polled through the public API five times a second, and the Binance USD-M books for the same assets, reconstructed from the exchange's 100 ms depth stream and periodic snapshots. The record runs from 15:00 UTC on 23 September to 20:00 UTC on 23 October, thirty days. Book measures are taken on a five-second grid: the quoted spread in basis points of the mid; depth, the dollar value of all quotes within 10 bps

Table 5: Volume, open interest, and turnover by contract and venue

	BTC	ETH	Gold	Silver
Kalshi				
Mean open interest ($m)	6.5	2.3	2.0	0.6
Mean daily volume ($m)	246	183	14	14
Median turnover (×)	32.8	60.7	5.1	19.2
Binance				
Mean open interest ($m)	7,119	4,616	422	201
Mean daily volume ($m)	10,370	7,952	1,584	612
Median turnover (×)	1.3	1.6	4.7	3.5
Kalshi / Binance turnover	24.5	38.2	1.1	5.5

Turnover is daily traded notional divided by the day's mean open interest; the table reports its median over the sample. Levels are sample means. Kalshi open interest and volume are from the margin-market candlestick history; Binance open interest is from the exchange's open-interest metrics and volume from the USD-M aggregated-trade archive. All figures in US dollars. BTC and ETH cover 3 June–20 September 2026 (110 days); gold and silver 10–20 September 2026 (11 days).

of the mid on both sides; the cost of buying $50,000 and $250,000 up the ask, as the average fill price's distance from the mid; and the share of ask depth held by the single largest price level. The window begins three days after the public analysis of 20 September and after the end of the trade sample, so it describes the venue while it was under scrutiny, and it includes 13 October, when a volume-based incentive program ended.

8.1 Depth and the cost of trading

Table 6 reports medians. Kalshi's BTC book holds $6.4 million within 10 bps of the mid against $46.6 million on Binance, and its ETH book $2.7 million against $22.9 million: seven to nine times thinner. The quoted spread is 0.36 bps in BTC and 0.41 bps in ETH, 30 and 11 times Binance's. The gap is largest for size. Buying $50,000 costs 1.6 bps in BTC and 1.7 bps in ETH on Kalshi and 0.006 and 0.02 bps on Binance; at $250,000 the ratios are about 350 and 120 to one. A taker entering and leaving a $250,000 position on Kalshi pays 4.2–5.1 bps in the book alone, before fees, where the same round trip on Binance costs a few hundredths of a basis point.

These costs put the turnover of Section 7 in perspective. On the median day open interest changed hands 33 times in BTC and 61 times in ETH (Table 5). Suppose every change of hands were a taker crossing the Kalshi book at the half-spread and paying the 1.0–4.0 bps list fee that applied to non-members from 18 August. ETH takers would then pay 0.7–2.6% of open interest each day, and BTC takers 0.4–1.4%, for positions that are not held. Turnover on that scale is expensive for anyone who pays to cross the book. It is close to free only for members trading under the rebate, and free for a member on both sides of the trade.

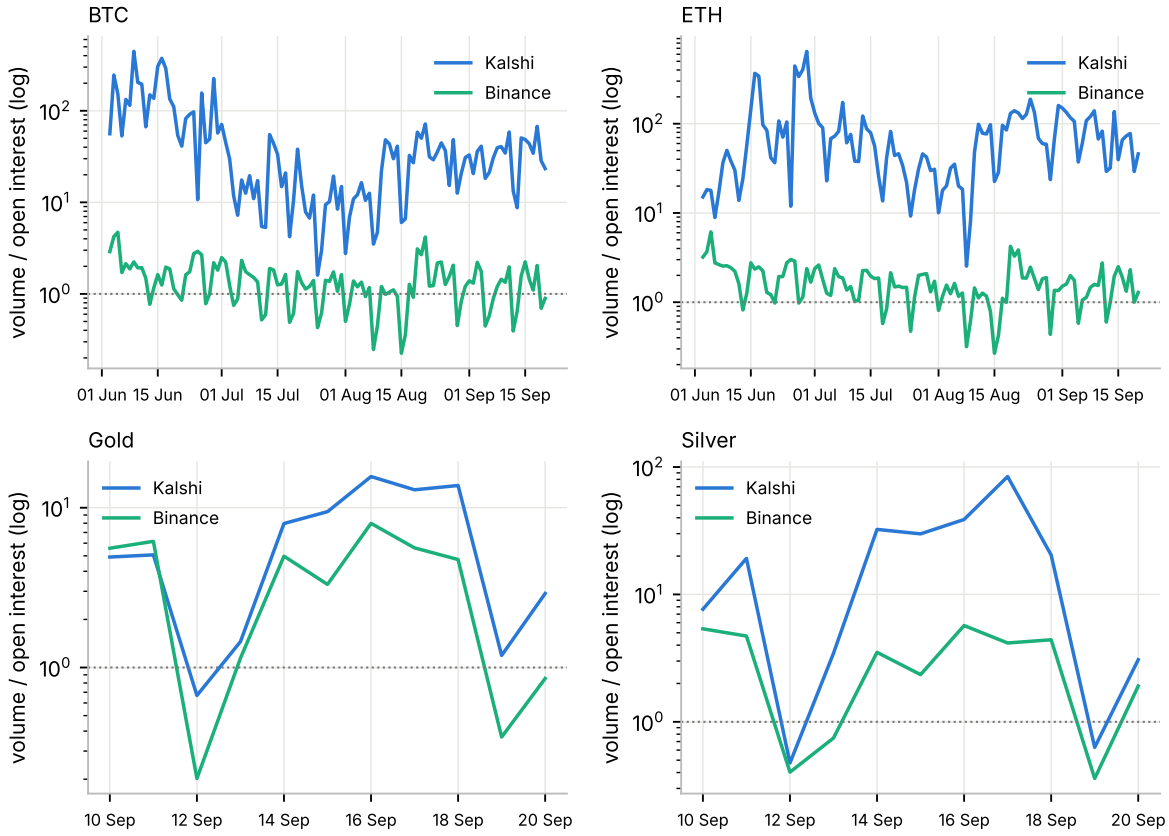


Figure 8: Daily turnover (traded notional divided by mean open interest, USD) for each contract on Kalshi (blue) and Binance USD-M (green), log scale. The dotted line marks a turnover of one. BTC and ETH span 3 June–20 September 2026; gold and silver span 10–20 September. The crypto contracts sit one to two orders of magnitude above the reference venue throughout, while gold tracks it closely, including the shared weekend troughs when COMEX is closed.

Table 6: Order-book liquidity, Kalshi and Binance, 23 September – 23 October 2026

		Spread	Depth ±10 bps	Cost to buy (bps)		Largest
Asset	Venue	(bps)	($m)	$50k	$250k	level
BTC	Kalshi	0.36	6.37	1.63	2.12	12%
	Binance	0.012	46.57	0.006	0.006	7%
ETH	Kalshi	0.41	2.69	1.74	2.53	17%
	Binance	0.039	22.89	0.020	0.021	6%
Gold	Kalshi	0.25	3.54	1.60	2.24	29%
	Binance	0.024	6.92	0.012	0.19	8%
Silver	Kalshi	1.48	3.46	2.08	2.89	30%
	Binance	1.65	3.59	0.83	1.11	10%

Medians over a five-second grid of order-book snapshots, 15:00 UTC 23 September to 20:00 UTC 23 October 2026. Spread and cost to buy are in basis points of the mid; cost to buy is the average fill price of a $50,000 or $250,000 market buy up the ask, relative to the mid. Depth is the dollar value of bid and ask quotes within 10 bps of the mid. Largest level is the share of ask depth within 2.5% of the mid held by the single largest price level. Kalshi books are polled five times a second through the public API; Binance USD-M books are reconstructed from the 100 ms depth stream and REST snapshots.

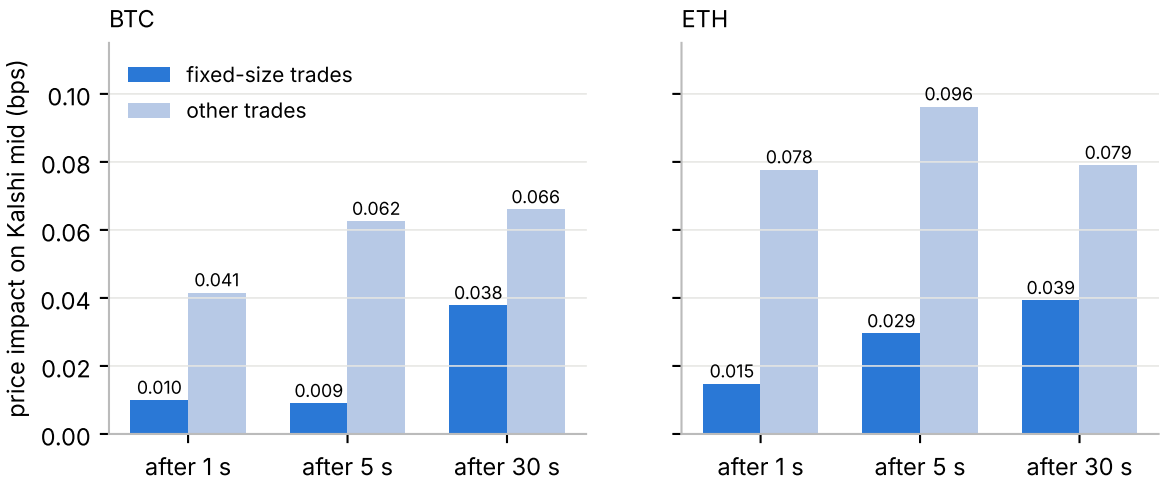


Figure 9: Price impact on the Kalshi mid, 23 September – 23 October 2026: twice the change in Kalshi's mid from the last book snapshot before a trade to 1, 5 and 30 seconds after, signed in the taker's direction, in basis points. Daily values averaged with weights equal to the day's trades. Fixed-size trades are trades within 2% of a dollar size that recurs at the top of the Kalshi book that day.

8.2 Fixed-size quotes in the book

The book shows directly what Section 4.3 inferred from trade timestamps. On the median snapshot the largest price level holds 12% of ask depth in Kalshi BTC and 17% in ETH, against 7% and 6% on Binance. The resting size at the top of the Kalshi book returns to a small set of dollar values day after day; in October the most frequent ETH values were about $4,000 and $5,200, beside a $4,050 ETH trade size that carried about half of ETH notional in the same weeks. Over the window, trades whose dollar size is within 2% of one of the day's recurring resting sizes are 58% of ETH trades and 23% of BTC trades.

8.3 Price impact

Kalshi's public response to the 20 September analysis attributed the flow to traders picking off quotes that had gone stale after a price move on another exchange (Kalshi, 2026). A trade that picks off a stale quote carries information: the quote it hits is replaced at a new price, and Kalshi's mid moves in the direction of the trade. I measure price impact as twice the change in Kalshi's own mid from the last snapshot before a trade to 1, 5 and 30 seconds after, signed in the taker's direction, and compare trades at a recurring resting size with all others (Figure 9).

Fixed-size trades move the mid by 0.010 bps after one second in BTC and 0.015 bps in ETH, a quarter and a fifth of the 0.041 and 0.078 bps moved by other trades in the same markets. After thirty seconds the figures are 0.038 against 0.066 bps in BTC and 0.039 against 0.079 bps in ETH. The fixed-size trades mostly hit a quote that is replaced where it was. This matches the taker markouts of Section 5.2, a gross edge against Binance of a few tenths of a basis point, and it is not what a flow made of stale-quote arbitrage looks like from the venue's own book.

8.4 The metals books

The metals contracts sharpen the comparison. Kalshi's gold and silver books are no deeper than its crypto books: $3.5 million within 10 bps of the mid in each, a $50,000 buy costing 1.6 and 2.1 bps, and close to a third of ask depth in a single price level (Table 6). Yet gold turned over 5.1 times its open interest over 10–20 September, in line with Binance (Table 5), and on 5 October Kalshi's API reported 24-hour volume of 8 times open interest in gold and 7.5 times in silver, against 35 times in BTC and 67 times in ETH. A thin book does not by itself produce extreme turnover on this venue. The crypto contracts' turnover comes with the fixed-size flow.

8.5 13 October: a volume incentive ends

On 28 September Kalshi filed the termination of its Volume Incentive Program, effective no earlier than 13 October (KalshiEX LLC, 2026e). The program paid members a share of a fixed reward per market in proportion to their volume. Its terms exempted perpetual futures from the price range that limited eligible trades in event contracts, and it excluded members with a Market Maker Agreement. The Liquidity Provider Program, which pays members with such an agreement for quoting obligations under terms filed as a confidential appendix, runs to 1 December 2027 (KalshiEX LLC, 2026d), and the perpetual fee rebate to the end of 2026. If the fixed-size flow were produced for the volume reward, it should have changed on 13 October.

It did not. In the week from 13 October ETH traded $338 million a day against $227 million in the week before, and its most common trade size, $4,050, carried 52% of notional against 45%. In BTC the most common size, $3,050, carried 29% against 28%.[5] The book did not move either. Kalshi's median daily cost of buying $50,000 was 1.69 bps in BTC and 1.74 bps in ETH from 13 October, against 1.58 and 1.71 bps before, and depth within 10 bps was $6.8 million and $2.8 million against $6.1 million and $2.8 million (Figure 10). The fixed-size flow does not depend on a volume reward open only to members without a market-maker agreement.

9 Interpretation

The fee-event evidence of Section 5, the turnover comparison of Section 7 and the order book of Section 8 agree that the fixed-size flow does not behave like trading and builds no position; the open question is who produces it. Three hypotheses are on the table. Under H_1 the maker and the taker have the same owner. Under H_2 they are two parties acting under an arrangement, for example a market-making contract with volume conditions. Under H_3 they are independent: a subsidized maker posts fixed-dollar quotes and an independent arbitrageur takes them.

The evidence against H_3 is circumstantial but consistent. The maker lost the full markout on every fill for eight weeks before any rebate could compensate it, and the taker's edge is at or below its fee since. The taker's order size tracked the maker's size through three changes, including a same-day change in two contracts. The taker did not become more selective when its fee rose from zero to its entire gross edge, while a separately identifiable population of

[5] Here a trade's size is its dollar notional rounded to $50, and the share is that of the most common size above $500. This rule needs no window below a round number. That matters because after 21 September the tickets moved to sizes such as $4,050 and $3,050, which the round-dollar rule of Section 3 does not cover.

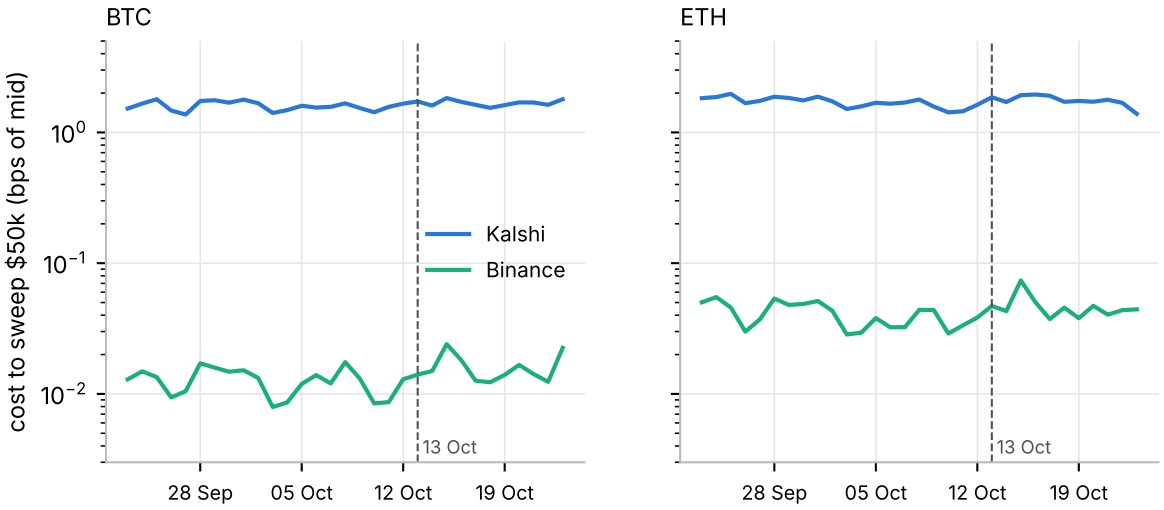


Figure 10: Cost of buying $50,000 up the ask, daily median of hourly averages, Kalshi (blue) and Binance (green), 23 September – 23 October 2026, log scale. Dashed line: 13 October, the earliest effective date of the Volume Incentive Program's termination.

arbitrageurs on the same venue reacted strongly to a fee cut the same day. The flow scaled up three- to fivefold within 42 hours of the fee switch-on once a full rebate was assured, and rose tenfold on 10 August without any public change in incentives. Its order sizes were changed in both contracts within ten seconds on 24 August and its quoting cycle in both contracts within minutes on 25 August. The order book recorded after the sample adds that the fixed-size trades barely move the venue's own price, a fifth to a half as much as other trades, which is not how stale-quote arbitrage looks. None of this is impossible for two independent firms, but each step requires a coincidence.

The evidence does not separate H_1 from H_2. The 3 June market-maker amendment for perpetuals is confidential, so a volume condition or a payment per contract cannot be ruled in or out. The fee schedule itself gives one public motive: until 24 June, volume set each member's initial fee tier, and maker tiers can be reached by holding a share of the venue's maker volume; a maker with half of all volume qualifies for the top tier on that ground alone. After 24 June the tier stopped mattering for members, because everything was rebated. One public program can be set aside: the flow and the book were unchanged when the Volume Incentive Program, which excluded members with a market-maker agreement, ended on 13 October. Whatever arrangement there is, if any, it does not run through that program.

What the evidence does establish is that between a fifth and two thirds of the volume reported for these contracts, depending on the week, is produced by a process that does not respond to its own trading profit, leaves no position, and costs nothing to run under the exchange's own rebate design. That is enough to say that the volume statistic does not measure trading interest. It is not enough to say who is on the other side.

10 Conclusion

Kalshi's perpetual futures have carried, since their first week, a flow of fixed-dollar quotes taken in full by same-sized orders that now accounts for about half of reported volume. The exchange's own filings show that this flow has been costless to produce for members throughout: no fees for three weeks, then a full rebate, then a split rebate that can sum up to zero. The flow scaled with the rebate, ignored a fee change that wiped out the taker's edge, and changed its size in both contracts within ten seconds and its cycle time in both contracts within minutes, on consecutive days. The gold and silver contracts listed under the same program show none of this and follow their reference market's hours. Order books recorded after the sample show the flow on a book seven to nine times thinner than Binance's, barely moving the venue's own price, and unchanged when a volume-based reward ended. Whether the two sides of the flow are one firm or two is not observable from public data; that they are not trading is.

These findings carry implications beyond the measurement of one venue's activity. Kalshi's management has publicly stated that the company is considering an initial public offering, with press reports of a targeted valuation near $40 billion against an annualized revenue run-rate of roughly $2 billion, while ruling out a listing in 2026 (CNBC, 2026). Because the exchange earns transaction-based revenue, a valuation of this order rests substantially on the level and growth of trading across both its event and its perpetual markets. To the extent that a material share of reported perpetual volume is manufactured rather than organic—the fixed-size flow documented here is roughly half of the notional in the BTC and ETH contracts, which are themselves the large majority of perpetual volume—headline volume and its trajectory may overstate the underlying trading demand on which such a valuation would rest. This distinction matters most for retail investors, who are the natural buyers of a prediction-market exchange at a public listing, are least able to audit trade-level data, and are therefore least equipped to discount inflated volume statistics and most exposed if a valuation anchored on them proves fragile. The evidence here does not extend to the event-contract business, which is most of the venue's activity and is not examined in this paper. It does show that reported volume on the perpetual complex cannot be taken at face value, which is the first input any outside assessment of the trading franchise requires.

A Elasticity of ticket and non-ticket flow to market activity

Table 7 reports the stacked regressions discussed in Section 5.3. Each column is a sample and each panel a Binance variable; the difference row is the coefficient on the interaction of that variable with the ticket indicator.

Table 7: Stacked elasticity regressions with a ticket interaction

	All	No fees 3–23 Jun	Full rebate 24 Jun–18 Aug	Taker 0.3 bps 19 Aug–20 Sep
Panel A: log Binance notional				
BTC				
Non-ticket flow	0.881	0.035	0.961	0.769
	(0.092)	(0.248)	(0.094)	(0.041)
Ticket flow	0.857	0.014	0.885	0.598
	(0.128)	(0.193)	(0.172)	(0.066)
Difference	-0.024	-0.021	-0.077	-0.171***
	(0.070)	(0.097)	(0.107)	(0.064)
Observations	5168	994	2604	1570
ETH				
Non-ticket flow	0.649	-0.297	0.821	0.752
	(0.143)	(0.323)	(0.129)	(0.041)
Ticket flow	0.465	-0.502	0.548	0.374
	(0.154)	(0.250)	(0.169)	(0.097)
Difference	-0.184***	-0.205	-0.273***	-0.378***
	(0.065)	(0.164)	(0.079)	(0.073)
Observations	5137	954	2619	1564
Panel B: log realized volatility				
BTC				
Non-ticket flow	1.088	-0.265	1.152	1.106
	(0.154)	(0.391)	(0.154)	(0.072)
Ticket flow	0.902	-0.301	0.954	0.766
	(0.200)	(0.315)	(0.269)	(0.114)
Difference	-0.187*	-0.036	-0.198	-0.340***
	(0.096)	(0.177)	(0.151)	(0.104)
Observations	5168	994	2604	1570
ETH				
Non-ticket flow	0.531	-0.999	1.058	1.120
	(0.281)	(0.503)	(0.218)	(0.060)
Ticket flow	0.286	-1.238	0.826	0.519
	(0.294)	(0.377)	(0.298)	(0.130)
Difference	-0.245**	-0.239	-0.232	-0.601***
	(0.114)	(0.270)	(0.152)	(0.096)
Observations	5137	954	2619	1564

OLS on a panel of contract-hours stacked over the two flow types (ticket and non-ticket), estimated separately by contract and sample. The dependent variable is log hourly Kalshi notional of the flow type. Regressors are the log of the Binance USD-M variable in the same asset (hourly notional in Panel A, realized volatility from one-minute returns in Panel B), an indicator for ticket flow, their interaction, and hour-of-day fixed effects. The non-ticket row is the coefficient on the Binance variable, the ticket row is the sum of that coefficient and the interaction, and the difference row is the interaction. Standard errors clustered by day in parentheses. *, **, ***: the difference is significant at 10, 5 and 1 percent.

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